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Inventor(s)

SAITO; Takuya

INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING SYSTEM, AND RECORDING MEDIUM

Abstract

Disclosed is an information processing apparatus that is configured to be communicably connected to a first terminal used by a learning instructor or an administrator and a plurality of second terminals used by a plurality of learners, the second terminals each including an electronic sticky note function for creating an electronic sticky note related to learning, and includes a processor that: acquires creation history information of the electronic sticky note created by the learners via the second terminals; in response to a school being designated as an analysis unit of usage status of the electronic sticky note function by the learning instructor or the administrator via the first terminal, derives a total number of the electronic sticky note created by the learners of a school associated with the learning instructor or the administrator based on the creation history information, and causes the first terminal to display a derived result.

Inventors: SAITO; Takuya (Tokyo, JP)

Applicant: CASIO COMPUTER CO., LTD. (Tokyo, JP)

Family ID: 93037427

Assignee: CASIO COMPUTER CO., LTD. (Tokyo, JP)

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-029752, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an information processing apparatus, an information processing system and a recording medium.

DESCRIPTION OF RELATED ART

[0003] There have been education support systems that analyze a course by the number of sticky notes for each page of contents related to the course and displays a course analysis result screen (for example, see JP 2019-212238A).

SUMMARY OF THE INVENTION

[0004] According to one aspect of the present disclosure, an information processing apparatus that is configured to be communicably connected to a first terminal configured to be used by a learning instructor or an administrator and a plurality of second terminals configured to be used by a plurality of learners, the plurality of second terminals each including an electronic sticky note function for creating an electronic sticky note related to learning, comprises [0005] a processor that: [0006] acquires creation history information of the electronic sticky note created by the plurality of learners via the plurality of second terminals; and [0007] in response to a school being designated as an analysis unit of usage status of the electronic sticky note function by the learning instructor or the administrator via the first terminal, derives a total number of the electronic sticky note created by the plurality of learners of a same school as a school associated with the learning instructor or the administrator based on the creation history information, and causes the first terminal to display a derived result.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] In the drawings:

[0009] FIG. 1 is a diagram of a schematic configuration of an information processing system according to the present embodiment;

[0010] FIG. 2 is a diagram of an example of the configuration of a database stored in a server;

[0011] FIG. 3 is a diagram of an example of a screen for a student to select an electronic dictionary to use;

[0012] FIG. 4 is a diagram of an example of a screen for a student to select a notebook to use;

[0013] FIG. 5 is a diagram of an example of a screen displayed when a notebook to be used is selected;

[0014] FIG. 6 is a flowchart illustrating a flow of an analysis process executed by a controller of the server;

[0015] FIG. 7 is a diagram of an example of a screen for selecting an item to be analyzed displayed when a teacher or an administrator logs in;

[0016] FIG. 8A is a diagram of an example of an analysis screen displayed when an electronic

sticky note function is selected as the item to be analyzed;
[0017] FIG. **8B** is a diagram of an example of an analysis screen displayed when the electronic sticky note function is selected as the item to be analyzed;
[0018] FIG. **8C** is a diagram of an example of an analysis screen displayed when the electronic sticky note function is selected as the item to be analyzed;
[0019] FIG. **9A** is a diagram of an example of an analysis screen displayed when an electronic dictionary function is selected as the item to be analyzed;
[0020] FIG. **9B** is a diagram of an example of an analysis screen displayed when the electronic dictionary function is selected as the item to be analyzed;
[0021] FIG. **10A** is a diagram of an example of an analysis screen displayed when a notebook function is selected as the item to be analyzed;
[0022] FIG. **10B** is a diagram of an example of an analysis screen displayed when the notebook function is selected as the item to be analyzed;
[0023] FIG. **10C** is a diagram of an example of an analysis screen displayed when the notebook function is selected as the item to be analyzed;
[0024] FIG. **11** is a diagram of an example of an analysis screen displayed when the number of accesses is selected as the item to be analyzed;
[0025] FIG. **12** is a diagram of an example of a graph displayed when the electronic sticky note function is selected as the item to be analyzed and “Homeroom” is selected as an analysis unit;
[0026] FIG. **13** is a diagram of an example of a graph displayed when the electronic sticky note function is selected as the item to be analyzed and “School” is selected as the analysis unit;
[0027] FIG. **14** is a diagram of an example of a graph displayed when the electronic dictionary function is selected as the item to be analyzed and “Homeroom” is selected as the analysis unit;
[0028] FIG. **15** is a diagram of an example of a graph displayed when the notebook function is selected as the item to be analyzed and “Homeroom” is selected as the analysis unit;
[0029] FIG. **16** is a diagram of an example of a graph displayed when the notebook function is selected as the item to be analyzed and “Course” is selected as the analysis unit; and
[0030] FIG. **17** is a diagram showing an example of a graph displayed when the number of accesses is selected as the item to be analyzed.

DETAILED DESCRIPTION

[0031] Hereinafter, an information processing system according to an embodiment of the present disclosure will be described in detail with reference to the drawings. It is assumed that the information processing system is used by the administrators of schools, teachers at the schools who are learning instructors, and students who are learners. The schools include not only regular elementary schools, junior high schools, high schools, and universities, but also vocational schools, cram schools, preparatory schools, and the like.

[0032] As shown in FIG. **1**, the information processing system **1** includes a server **10**, which is an information processing apparatus, a first terminal **20A**, and a plurality of second terminals **20B**. The server **10**, the first terminal **20A**, and the second terminals **20B** are communicatively connected via a network **30**. The network **30** is, for example, an arbitrary communication network such as the Internet, a wireless local area network (LAN), a wired LAN, a mobile communication network, a short-distance wireless communication network, or a combination of some or all of these. The information processing system **1** may include a plurality of first terminals **20A**.

[0033] The server **10** is a web server that provides a learning support service to the first terminal **20A** and second terminals **20B** via the web. As shown in FIG. **1**, the server **10** includes a controller (processor) **11**, a storage **12**, and a communicator **13**. The controller **11**, storage **12**, and communicator **13** are connected to each other via a system bus **19**.

[0034] The controller **11** includes a central processing unit (CPU), a random-access memory (RAM), and the like. The CPU of the controller **11** is a processor that controls behavior of each unit of the server **10** by reading and executing programs stored in the storage **12** and performing various

arithmetic processing. The controller **11** may include a plurality of processors, for example, a plurality of CPUs. The plurality of processors may execute a plurality of processes executed by the controller **11** of the present embodiment. In this case, the plurality of processors may be involved in a common process. Alternatively, the plurality of processors may independently execute different processes in parallel.

[0035] The storage **12** is a non-transitory recording medium readable by the CPU of the controller **11**. The storage **12** stores programs and various data. The storage **12** includes a nonvolatile memory such as a hard disk drive (HDD) or a solid-state drive (SSD). As shown in FIG. 2, the storage **12** stores, for example, a school database **121**, a homeroom database **122**, a course database **123**, a user database **124**, a notebook database **125**, a sticky note database **126**, a dictionary database **127**, an operation history database **128**, an access history database **129**, and the like.

[0036] The school database **121** stores information about schools using the learning support service provided by the information processing system **1**. The school database **121** includes a table that stores information such as a school ID and a school name in association with each of the schools using the learning support service. The school ID is identification information for uniquely identifying each of the schools that uses the learning support service. The school name is the name of the school.

[0037] The homeroom database **122** stores information about homerooms of each school using the learning support service. The homeroom database **122** includes a table that stores information such as a homeroom ID, a homeroom name, and a school ID in association with each of the homerooms of each school using the learning support service. The homeroom ID is identification information for uniquely identifying each homeroom of each school that uses the learning support service. The homeroom name is the name of the homeroom. Examples of the homeroom name include “Homeroom 1, Grade 7”. The school ID is the school ID of the school to which the homeroom belongs.

[0038] The course database **123** stores information about courses conducted at each school using the learning support service. The course database **123** includes a table that stores information such as a course ID, a course name, a subject, a teacher ID, and a school ID in association with each of the courses conducted at each school using the learning support service. The course ID is identification information for uniquely identifying each course of each school using the learning support service. The course name is the name of the course. Examples of the course name include a name that combines a homeroom and a subject such as “Mathematics, Homeroom 1, Grade 7”, a name of an elective subject such as “Elective Chemistry”, and the like. The subject is the name of the subject of the course. The teacher ID is the user ID of the teacher who teaches the course. The school ID is the school ID of the school at which the course is conducted.

[0039] The user database **124** stores information about users using the learning support service. The user database **124** includes a table that stores information such as a user ID, a password, a user category, a user name, a school ID, a homeroom ID, and a course ID in association with each of the users using the learning support service. The user ID is identification information for uniquely identifying each of the users using the learning support service. The user category indicates whether a user is the administrator, a teacher, or a student. The school ID is the school ID of the school to which the user belongs. The homeroom ID is the homeroom ID of the homeroom to which the user belongs when the user is a teacher or a student. The course ID is the course ID of the course to which the user belongs. When the user is a teacher, the course to which the user belongs is a course that the user teaches. When the user is a student, the course to which the user belongs is a course that the user is taking. When the user is a teacher or a student, one homeroom is set for the user. When the user is a teacher or a student, a plurality of courses is set for the user. One school is set for each user. Now, the homeroom falls under a first classification and a second classification in the present disclosure. The course falls under the first classification and the second classification in the present disclosure. The school falls under the first classification in the present

disclosure. That is, the users are grouped into a first group belonging to the first classification corresponding to the first learning function described later, and a second group belonging to the second classification corresponding to the second learning function described later. The number of persons applicable to each of the analysis units is basically as follows: $\text{homeroom} \leq \text{course} < \text{school}$. In some cases, there may be $\text{homeroom} > \text{course}$.

[0040] The notebook database **125** stores information about notebooks, which are electronic notebooks, created in the learning support service. In association with each of the notebooks created in the learning support service, the notebook database **125** includes a table that stores information such as a notebook ID, a notebook name, a user ID, a course ID, a thumbnail image of the notebook, and the creation date and time. The notebook ID is identification information for uniquely identifying the created notebook. The user ID is the user ID of the user who has created the notebook. The course ID is the course ID of the course for which the notebook has been created.

[0041] The sticky note database **126** stores information about electronic sticky notes created in the learning support service. In association with each of the electronic sticky notes created in the learning support service, the sticky note database **126** includes a table that stores information such as a sticky note ID, a user ID, a notebook ID, a type of the sticky note, contents of the sticky note, a thumbnail image of the sticky note, and the creation date and time. The sticky note ID is identification information for uniquely identifying the created sticky note. The notebook ID is the notebook ID of the notebook in which the sticky note has been created. The type of the sticky note is information indicating the type of the sticky note such as a text sticky note, a link sticky note, a file sticky note, a dictionary sticky note, or a math sticky note.

[0042] The dictionary database **127** stores information about electronic dictionaries usable by the user in the learning support service. In association with each of the electronic dictionaries usable by the user in the learning support service, the dictionary database **127** includes a table that stores information such as a dictionary ID, a dictionary name, a type, a storage location, and a thumbnail image of the dictionary. The dictionary ID is identification information for uniquely identifying the electronic dictionary available in the learning support service. The dictionary name is the name of the electronic dictionary. The type is information indicating the type of the dictionary such as an English-Japanese dictionary or a Japanese dictionary. The storage location is information indicating the location where data of the electronic dictionary is stored. The thumbnail image of the dictionary is image data of a thumbnail image such as the front cover of the dictionary.

[0043] The operation history database **128** stores information about operation histories of the users in the learning support service. The operation history database **128** includes a table that stores information such as a user ID, an operation ID, a date and time, a dictionary ID, and a course ID in association with each operation. The user ID is the user ID of the user who has performed the operation. The operation ID is identification information for identifying the operation performed by the user. The date and time are the date and time when the operation has been performed. The dictionary ID is identification information, when an operation related to a dictionary is performed, for identifying the operated dictionary. The course ID is identification information, when an operation related to a notebook is performed, for identifying the course in which the operation has been performed.

[0044] The access history database **129** stores an access history when screens on the website providing the learning support service are accessed. The access history database **129** includes a table that stores information such as a user ID, a screen ID, and a date and time in association with each access to a screen on the website. The user ID is the user ID of the user who has accessed the screen. The screen ID is identification information of the accessed screen. The date and time are the date and time of the access.

[0045] In the storage **12**, at least one analysis unit among the homeroom, course, and school is stored in association with each learning function provided by the server **10** as an analysis unit of

the usage status of the learning function. The first learning function, which will be described later, is associated with the analysis units of the homeroom, course, and school, which fall under the first classification. The second learning function, which will be described later, is associated with the analysis units of the homeroom and course, which fall under the second classification. An analysis unit not falling under the first classification may be associated with the first learning function. Also, an analysis unit not falling under the second classification may be associated with the second learning function. The communicator **13** performs a communication operation in accordance with a predetermined communication standard. By this communication operation, the communicator **13** transmits and receives data to and from the first terminal **20A** or each of the second terminals **20B** via the network **30**.

[0046] The first terminal **20A** is a terminal device used by the administrator or a teacher. The first terminal **20A** may be any of a personal computer, a portable terminal device such as a tablet, and a mobile communication device such as a smartphone. The first terminal **20A** includes a controller **21A**, a storage **22A**, an operation unit **23A**, a display **24A**, a communicator **25A**, and an image capturer **27A**. The controller **21A**, storage **22A**, operation unit **23A**, display **24A**, communicator **25A**, and image capturer **27A** are connected to each other via a system bus **29A**.

[0047] The controller **21A** includes a CPU, a RAM, and the like. The CPU of the controller **21A** is a processor that controls behavior of each unit of the first terminal **20A** by reading and executing a program stored in the storage **22A** and performing various arithmetic processing. The controller **21A** may include a plurality of processors, for example, a plurality of CPUs. The plurality of processors may execute a plurality of processes executed by the controller **21A** of the present embodiment. In this case, the plurality of processors may be involved in a common process.

Alternatively, the plurality of processors may independently execute different processes in parallel. [0048] The storage **22A** is a non-transitory recording medium readable by the CPU of the controller **21A**. The storage **22A** stores a program and various data. The storage **22A** includes, for example, a nonvolatile memory such as an HDD or an SSD. The program stored in the storage **22A** includes a web browser. The operation unit **23A** includes, for example, at least one of a keyboard, a touch panel, or a mouse. The operation unit **23A** receives an operation by the user, and outputs an input signal corresponding to the operation to the controller **21A**.

[0049] The display **24A** includes a liquid crystal display (LCD) or an electro luminescence (EL) display. The display **24A** performs various displays according to display information instructed by the controller **21A**. The communicator **25A** performs a communication operation in accordance with a predetermined communication standard. By this communication operation, the communicator **25A** transmits and receives data to and from the server **10** via the network **30**.

[0050] The image capturer **27A** is a digital camera that includes an optical system and an imaging element such as a charge coupled device (CCD) or a complementary metal oxide semiconductor (CMOS). The image capturer **27A** captures an image of a subject, generates image data of a still image or a moving image, and outputs the image data to the controller **21A**.

[0051] Each of the second terminals **20B** is a terminal device used by a student. Similarly to the first terminal **20A**, the second terminal **20B** may be any of a personal computer, a portable terminal device such as a tablet, and a mobile communication device such as a smartphone. The second terminal **20B** includes a controller **21B**, a storage **22B**, an operation unit **23B**, a display **24B**, a communicator **25B**, and an image capturer **27B**. The controller **21B**, storage **22B**, operation unit **23B**, display **24B**, communicator **25B**, and imaging capture **27B** are connected to each other via a system bus **29B**.

[0052] The controller **21B** includes a CPU, a RAM, and the like. The CPU of the controller **21B** is a processor that controls behavior of each unit of the second terminal **20B** by reading and executing a program stored in the storage **22B** and performing various arithmetic processing. The controller **21B** may include a plurality of processors, for example, a plurality of CPUs. The plurality of processors may execute a plurality of processes executed by the controller **21B** of the present

embodiment. In this case, the plurality of processors may be involved in a common process. Alternatively, the plurality of processors may independently execute different processes in parallel. The storage **22B**, operation unit **23B**, display **24B**, communicator **25B**, and image capturer **27B** have the same configurations as the storage **22A**, operation unit **23A**, display **24A**, communicator **25A**, and image capturer **27A**, respectively, and thus the description thereof is omitted.

[0053] Next, behavior of the information processing system **1** will be described. First, described is a process executed by the controller **11** of the server **10** when a student logs in to the website of the learning support service via a second terminal **20B**.

[0054] When the student logs in to the website of the learning support service via the second terminal **20B**, the controller **11** of the server **10** provides the learning support service for students to the second terminal **20B** in cooperation with the program stored in the storage **12**. The learning support service for students provides a plurality of different learning functions to the second terminal **20B**. The learning functions provided by the learning support service for students include an electronic dictionary function, a notebook function, and an electronic sticky note function. The notebook function and the electronic sticky note function correspond to the first learning function of the present disclosure. The electronic dictionary function corresponds to the second learning function of the present disclosure. The controller **11** performs the following processing to provide the plurality of different learning functions.

[0055] When the student logs in to the web site of the learning support service via the second terminal **20B**, the controller **11** of the server **10** generates screen information for displaying a screen **241** by the browser for the student to select a learning function to use. Then, the controller **11** transmits the generated screen information to the second terminal **20B** via the communicator **13**. Thus, the controller **11** causes the display **24B** of the second terminal **20B** to display the screen **241**.

[0056] The screen **241** includes a region **2411** and a region **2412**, as shown in FIG. **3**. In the region **2411**, a selection field for the user to select a learning function to be used is displayed. In the selection field, for example, a dictionary content selection field **241a** and a notebook selection field **241b** are displayed. The dictionary content selection field **241a** is a field for selecting an electronic dictionary function as a learning function to be used. The notebook selection field **241b** is a field for selecting a notebook function as a learning function to be used. In the notebook selection field **241b**, a selection field for selecting a course that uses notebooks is displayed. In the region **2412**, a user interface (UI) for using the learning function selected in the region **2411** is displayed. In the present embodiment, the electronic dictionary function is selected by default, and a list for the user to select an electronic dictionary to use is displayed in the region **2412**.

[0057] For example, when the log-in is completed, the controller **11** reads the thumbnail images of the respective electronic dictionaries stored in the dictionary database **127**, generates screen information for displaying the screen **241** in which the thumbnail images of the electronic dictionaries are displayed in a list in the region **2412**, and transmits the generated screen information to the second terminal device **20B** via the communicator **13**.

[0058] When the second terminal **20B** receives the screen information of the screen **241** from the server **10** via the communicator **25B**, the controller **21B** causes the display **24B** to display the screen **241** in cooperation with the browser. When one of the thumbnail images of the electronic dictionaries is selected by an operation via the operation unit **23B** on the displayed screen **241**, the controller **21B** transmits operation information to the server **10** via the communicator **25B** indicating that the operation of selecting the electronic dictionary has been performed. The operation information includes, for example, information such as a user ID, an operation ID, a dictionary ID, and the operation date and time. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID indicating that the operation of selecting an electronic dictionary has been performed. The dictionary ID is the dictionary ID of the selected electronic dictionary.

[0059] When the server **10** receives the operation information from the second terminal **20B** via the communicator **13** indicating that the operation of selecting the electronic dictionary has been performed, the controller **11** causes the display **24B** of the second terminal **20B** to display the contents of the selected electronic dictionary. For example, the controller **11** reads the data of the selected electronic dictionary from the dictionary database **127**, generates screen information for displaying the contents of the selected electronic dictionary, and transmits the generated screen information to the second terminal **20B** via the communicator **13**.

[0060] When the second terminal **20B** receives the screen information of the contents of the electronic dictionary transmitted from the server **10** via the communicator **25B**, the controller **21B** causes the display **24B** to display the contents of the electronic dictionary in cooperation with the browser. When an operation such as a word search is performed from the displayed contents, the controller **21B** transmits operation information indicating the performed operation to the server **10** via the communicator **25B**. The operation information includes, for example, information such as a user ID, an operation ID, a dictionary ID, and the operation date and time. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID indicating the performed operation. The dictionary ID is the dictionary ID of the electronic dictionary in use.

[0061] When the server **10** receives the operation information transmitted from the second terminal **20B** via the communicator **13** indicating the operation performed on the contents of the electronic dictionary, the controller **11** causes the display **24B** of the second terminal **20B** to display a screen corresponding to the performed operation. For example, when the operation is to instruct a search for a word, the controller **11** searches for the word for which the search is instructed using the dictionary database **127** and generates screen information for displaying a screen showing the search result. Then, the controller **11** transmits the generated screen information to the second terminal **20B** via the communicator **13**. When the controller **11** receives the operation information transmitted from the second terminal **20B** indicating the operation performed on the contents of the electronic dictionary, the controller **11** adds a record to the operation history database **128** and stores the operation history based on the received operation information. For example, when an operation of searching for a word is performed, the controller **11** stores the following in the operation history database **128** as the operation history: the user ID of the logged-in student; the operation ID representing the search in the electronic dictionary; the search date and time; and the dictionary ID of the electronic dictionary used for the search. The screen **241** for using the electronic dictionary function is also displayed on the display **24B** of the second terminal **20B** by the same procedure as described above when the dictionary content selection field **241a** is pressed by an operation via the operation unit **23B**.

[0062] On the other hand, when an operation of selecting a notebook function of a course in the notebook selection field **241b** is performed on the screen **241** by an operation via the operation unit **23B**, the controller **21B** transmits operation information to the server **10** via the communicator **25B** indicating that the operation of selecting the notebook function has been performed. The operation information includes, for example, information such as a user ID, an operation ID, a course ID, and the operation date and time. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID indicating that the operation of selecting a notebook function has been performed. The course ID is the course ID of the selected course.

[0063] When the server **10** receives the operation information transmitted from the second terminal **20B** via the communicator **13** indicating that the operation of selecting the notebook function has been performed, the controller **11** generates screen information for displaying a screen **242** by the browser and transmits the generated screen information to the second terminal **20B** via the communicator **13**. Thus, the controller **11** causes the display **24B** of the second terminal **20B** to display the screen **242**. The screen **242** is a screen for the user to use the notebook function and the electronic sticky note function. In the present embodiment, the notebook function is to create a notebook and is provided on the notebook. The electronic sticky note function is to create an

electronic sticky note. Since an electronic sticky note is created on a notebook, the notebook function includes the electronic sticky note function.

[0064] The screen **242** includes a region **2421** and a region **2422**, as shown in FIG. **4**. The region **2421** is the same as the region **2411**, and an electronic dictionary selection filed **242a** and a notebook selection field **242b** are displayed in the region **2421**. In the region **2422**, a notebook list **242c** of the selected course and an operation button **242d** for creating a new notebook are displayed.

[0065] For example, the controller **11** searches the notebook database **125** with the user ID and the course ID included in the operation information transmitted from the second terminal **20B**. Next, the controller **11** acquires the thumbnail image of the notebook from each record obtained by the search. Then, as shown in FIG. **4**, the controller **11** generates screen information of the screen **242** in which the acquired thumbnail image is displayed on the notebook list **242c** and transmits the generated screen information to the second terminal **20B** via the communicator **13**.

[0066] When the second terminal **20B** receives the screen information for displaying the screen **242** transmitted from the server **10** via the communicator **25B**, the controller **21B** causes the display **24B** to display the screen **242** in cooperation with the browser. When the operation button **242d** on the screen **242** is pressed by an operation via the operation unit **23B**, the controller **21B** causes the display **24B** to display a screen for creating a new notebook as a pop-up. When a notebook name is input by an operation via the operation unit **23B** and an operation button for instructing to create a notebook is pressed from the pop-up screen, the controller **21B** transmits operation information to the server **10** via the communicator **25B** indicating that the operation for creating the notebook has been performed. The operation information includes information such as a user ID, an operation ID, the operation date and time, a course ID, and a notebook name. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID indicating that the operation of creating a notebook has been performed. The course ID is the course ID of the selected course. The notebook name is the input notebook name.

[0067] When the server **10** receives the operation information transmitted from the second terminal **20B** via communicator **13** indicating that the operation to create the notebook has been performed, the controller **11** issues a new notebook ID. The controller **11** also creates a thumbnail image in which the input notebook name and the like are displayed. Next, the controller **11** adds a record to the notebook database **125**, and stores the notebook name, the user ID, the course ID, the thumbnail image of the notebook, the creation date and time, and the like based on the issued notebook ID and the received operation information. Then, the controller **11** generates screen information of the screen **242** in which the thumbnail image of the new notebook is added to the notebook list **242c** and transmits the generated screen information to the second terminal **20B** via communicator **13**. Further, the controller **11** adds a record to the operation history database **128**, and stores in the operation history database **128** the user ID, the operation ID, the operation date and time, and the course ID included in the operation information.

[0068] When the second terminal **20B** receives the screen information transmitted from the server **10** via the communicator **25B** for displaying the screen **242** in which the thumbnail image of the new notebook is added, the controller **21B** causes the display **24B** to display the screen **242** in cooperation with the browser.

[0069] Further, in second terminal **20B**, when an operation of selecting one of the notebooks from the notebook list **242c** on the screen **242** is performed by an operation via the operation unit **23B**, the controller **21B** transmits operation information to the server **10** via the communicator **25B** indicating that the operation of selecting the notebook has been performed. The operation information includes a user ID, an operation ID, the operation date and time, a course ID, a notebook ID, and the like. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID indicating that the operation of selecting the notebook has been performed. The course ID is the course ID of the selected course. The notebook ID is the notebook ID of the

selected notebook.

[0070] When the server **10** receives the operation information transmitted from the second terminal **20B** via communicator **13** indicating that the operation of selecting the notebook has been performed, the controller **11** generates screen information of a screen **243** showing the contents of the selected notebook. Then, the controller **11** transmits the generated screen information to the second terminal **20B** via the communicator **13**. When the second terminal **20B** receives the screen information for displaying the screen **243** of the selected notebook transmitted from the server **10** via the communicator **25B**, the controller **21B** causes the display **24B** to display the screen **243** in cooperation with the browser.

[0071] On the screen **243**, as shown in FIG. 5, a text sticky note button **243a**, a camera sticky note button **243b**, a link sticky note button **243c**, a file sticky note button **243d**, a dictionary sticky note button **243e**, a math sticky note button **243f**, and the like are arranged. A sticky note region **243g** is also provided on the screen **243**. The sticky note region **243g** is for displaying an electronic sticky note created on the notebook. In the notebook, contents learned by the user and information acquired for learning are recorded as an electronic sticky note. If an electronic sticky note has already been created, a thumbnail image of the electronic sticky note is displayed in the sticky note region **243g**.

[0072] The text sticky note button **243a** is an operation button for instructing to create a text sticky note. The text sticky note is an electronic sticky note for recording and displaying information entered in text. The camera sticky note button **243b** is an operation button for instructing to create a camera sticky note. The camera sticky note is an electronic sticky note for recording and displaying an image captured by the image capturer **27B** or the like. The link sticky note button **243c** is an operation button for instructing to create a link sticky note. The link sticky note is an electronic sticky note for recording and displaying a uniform resource locator (URL) of a linked website. The file sticky note button **243d** is an operation button for instructing to create a file sticky note. The file sticky note is an electronic sticky note for recording and displaying an image file, a portable document format (PDF) file, or the like. The dictionary sticky button **243e** is an operation button for instructing to create a dictionary sticky note. The dictionary sticky note is an electronic sticky note for recording and displaying information searched in an electronic dictionary. The math sticky button **243f** is an operation button for instructing to create a math sticky note. The math sticky note is a sticky note for recording and displaying an entered mathematical expression, graph, or the like.

[0073] On the screen **243**, a store icon **243h**, a submit icon **243i**, and a send/receive icon **243j** are arranged. The store icon **243h** is an icon for instructing to store an electronic sticky note dragged onto the store icon **243h** in a predetermined storage destination. The submit icon **243i** is an icon for instructing to submit an electronic sticky note dragged onto the submit icon **243i** to a designated submission destination. The send/receive icon **243j** is an icon for instructing to transmit an electronic sticky note dragged onto the send/receive icon **243j** to a designated destination.

[0074] In the second terminal **20B**, when one of the text sticky note button **243a** to the math sticky note button **243f** on the screen **243** is operated by an operation via the operation unit **23B**, the controller **21B** causes the display **24B** to display a UI for creating an electronic sticky note according to the type of the operated button. The controller **21B** creates an electronic sticky note in response to the operation on the displayed UI, and displays the electronic sticky note in the sticky note region **243g**. Then, the controller **21B** transmits to the server **10** via the communicator **25B** operation information indicating that the operation of creating the electronic sticky note has been performed and electronic sticky note information which is content of information of the created electronic sticky note. The transmitted operation information includes a user ID, an operation ID, the operation date and time, a course ID, a notebook ID, and the like. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID indicating that the operation of creating the electronic sticky note has been performed. The operation ID is an ID that allows the type of the created electronic sticky note to be specified. The course ID is the course ID of the

selected course. The notebook ID is the notebook ID of the notebook in which the electronic sticky note is created.

[0075] When the server **10** receives from the second terminal **20B** via the communicator **13** the operation information indicating that the operation of creating the electronic sticky note has been performed and the electronic sticky note information, the controller **11** issues a new sticky note ID, and creates a thumbnail image of the created electronic sticky note. Next, the controller **11** adds a record to the sticky note database **126**, and stores the user ID, the notebook ID, the type of the sticky note, the content of the electronic sticky note, the thumbnail image, and the like based on the issued sticky note ID and the received operation information. The controller **11** also adds a record to the operation history database **128**, and stores the user ID, the operation ID, the operation date and time, and the like in the operation history database **128** as the operation history based on the received operation information. The information stored here corresponds to the creation history information of electronic sticky notes.

[0076] In the second terminal **20B**, when the store icon **243h**, submit icon **243i**, or send/receive icon **243j** of the screen **243** is operated by an operation via the operation unit **23B** and an operation instructing to store, submit, or send/receive an electronic sticky note is executed, the controller **21B** transmits operation information to the server **10** via the communicator **25B** indicating that the operation instructing to store, submit, or send/receive the electronic sticky note has been performed. The transmitted operation information includes a user ID, an operation ID, the operation date and time, a sticky note ID, a destination, and the like. The user ID is the user ID of the currently logged-in user. The operation ID is the operation ID of the operation instructing to store, submit, or send/receive the electronic sticky note. The sticky note ID is the sticky note ID of the electronic sticky note to be stored, submitted, or sent/received. The destination is information of a storage destination, a submission destination, or a transmission/reception destination of the electronic sticky note.

[0077] When the server **10** receives from the second terminal **20B** via the communicator **13** the operation information indicating that the operation instructing to store, submit, or send/receive the electronic sticky note has been performed, the controller **11** adds a record to the operation history database **128**, and stores the user ID, the operation ID, and the operation date and time in the operation history database **128** as the operation history. The controller **11** reads the electronic sticky note information of the received sticky note ID from the sticky note database **126**, and stores or transmits the read electronic sticky note information to the designated destination. Thus, when the student logs in from the second terminal **20B**, the controller **11** of the server **10** provides the electronic dictionary function, the notebook function, and the electronic sticky note function in response to an operation by the student.

[0078] In addition, each time a screen on the website that provides the learning support service is accessed via the first terminal **20A** or the second terminal **20B**, the controller **11** adds a record to the access history database **129** and stores a user ID, a screen ID, and a date and time. The user ID is the user ID of the user who has accessed the screen. The screen ID is the identification information of the accessed screen. The date and time are the date and time of the access.

[0079] Next, a process when the administrator or a teacher logs in to the learning support service via the first terminal **20A** will be described. When the administrator or the teacher logs in via the first terminal **20A** and performs a predetermined operation for instructing analysis of a learning function, the controller **11** of the server **10** executes the analysis process shown in FIG. 6.

Specifically, the controller **11** executes the analysis process in cooperation with a program stored in the storage **12**. The analysis process will be described below with reference to FIG. 6.

[0080] In the analysis process, first, the controller **11** causes the display **24A** of the first terminal **20A** to display a screen **341** for selecting an item to be analyzed (step **S1**). For example, the controller **11** generates screen information for displaying the screen **341** by the browser and transmits the generated screen information to the first terminal **20A** via the communication **13**.

Thus, the controller **11** causes the display **24A** of the first terminal **20A** to display the screen **341**. [0081] On the screen **341**, as shown in FIG. 7, an electronic sticky note selection button **341a**, an electronic dictionary selection button **341b**, a notebook selection button **341c**, and a number of accesses selection button **341d** are arranged. The electronic sticky note selection button **341a** is a button for selecting the electronic sticky note function as the item to be analyzed. The electronic dictionary selection button **341b** is a button for selecting the electronic dictionary function as the item to be analyzed. The notebook selection button **341c** is a button for selecting the notebook function as the item to be analyzed. The number of accesses selection button **341d** is a button for selecting the number of accesses as the item to be analyzed.

[0082] The controller **11** waits for reception of operation information transmitted from the first terminal **20A** via the communicator **13** indicating that an operation of selecting the item to be analyzed has been performed (step S2). When the first terminal **20A** receives the screen information of the screen **341** transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **341** in cooperation with the browser. When one of the selection buttons is selected from the displayed screen **341** by an operation via the operation unit **23A**, the controller **21A** transmits operation information to server **10** via the communicator **25A** indicating that the operation of selecting the item to be analyzed has been performed. The operation information includes information indicating the selected item.

[0083] When it is determined that the operation information transmitted from the first terminal **20A** indicating that the operation of selecting the item to be analyzed has been performed has been received (step S2; YES), the controller **11** causes display **24A** of the first terminal **20A** to display one of a screen **342** to a screen **345** for analysis (see FIG. 8A to FIG. 11) in accordance with the selected item (step S3). The controller **11** determines the classification corresponding to the analysis units of the selected item based on the information of the analysis units of each learning function stored in the storage **12**. Based on the determined classification, the controller **11**, for example, generates screen information for displaying one of the screen **342** to the screen **345** by the browser, and transmits the generated screen information to the first terminal **20A** via the communicator **13**. Thus, the controller **11** causes the display **24A** of the first terminal **20A** to display one of the screen **342** to the screen **345** shown in FIG. 8A to FIG. 11, respectively, in accordance with the selected item.

[0084] The screen **342** is displayed when the electronic sticky note function, that is, the first learning function is selected as the item to be analyzed, as shown in FIG. 8A to FIG. 8C. The screen **342** is for analyzing the usage status of the electronic sticky note function. As described above, the first learning function is associated with the homeroom, course, and school, which fall under the first classification, as the analysis units. Therefore, a radio button **342a** for selecting the homeroom, course, or school as the analysis unit is displayed on the screen **342**. When the homeroom is selected by an operation via the operation unit **23A** on the screen **342**, a pull-down **342b** for selecting a homeroom name to be analyzed is displayed as shown in FIG. 8A. When the course is selected by an operation via the operation unit **23A** on the screen **342**, a pull-down **342c** for selecting a teacher to be analyzed is displayed as shown in FIG. 8B. When the school is selected by an operation via the operation unit **23A** on the screen **342**, a selection field **342d** for selecting a period to be analyzed is displayed as shown in FIG. 8C.

[0085] The screen **343** is displayed when the electronic dictionary function, that is, the second learning function is selected as the item to be analyzed, as shown in FIG. 9A to FIG. 9B. The screen **343** is for analyzing the usage status of the electronic dictionary function. As described above, the second learning function is associated with the homeroom and course, which fall under the second classification, as the analysis units. Therefore, a radio button **343a** for selecting the homeroom or course as the analysis unit is displayed on the screen **343**. A radio button for selecting the school is not displayed on the screen **343**. The electronic dictionary function uses different dictionaries for each grade. Therefore, it is meaningless to analyze the usage status of each

electronic dictionary in the entire school. When the homeroom is selected by an operation via the operation unit **23A** on the screen **343**, a pull-down **343b** for selecting a homeroom name to be analyzed is displayed as shown in FIG. **9A**. When the course is selected by an operation via the operation unit **23A** on the screen **343**, a pull-down **343c** for selecting a teacher to be analyzed is displayed as shown in FIG. **9B**.

[0086] The screen **344** is displayed when the notebook function, that is, the first learning function is selected as the item to be analyzed, as shown in FIG. **10A** to FIG. **10C**. The screen **344** is for analyzing the usage status of the notebook function. Therefore, a radio button **344a** for selecting the homeroom, course, or school as the analysis unit is displayed on the screen **344**. When the homeroom is selected by an operation via the operation unit **23A** on the screen **344**, a pull-down **344b** for selecting a homeroom name to be analyzed is displayed as shown in FIG. **10A**. When the course is selected by an operation via the operation unit **23A** on the screen **344**, a pull-down **344c** for selecting a teacher to be analyzed is displayed as shown in FIG. **10B**. When the school is selected by an operation via the operation unit **23A** on the screen **344**, a selection field **344d** for selecting a period to be analyzed is displayed as shown in FIG. **10C**.

[0087] The screen **345** is displayed when the number of accesses is selected as the item to be analyzed, as shown in FIG. **11**. A selection field **345a** for selecting a period to be analyzed is displayed on the screen **345**. A radio button or the like for selecting an analysis unit is not displayed on the screen **345**. This is because the analysis of the number of accesses is a function for allowing the administrator to easily grasp the usage status of the learning support service in the entire school.

[0088] As described above, the controller **11** changes the radio button for selecting the analysis unit for each item selected as the analysis target of the usage status to display the screen for analysis. The controller **11** also displays the screen for analysis with the pull-down items changing in accordance with the analysis unit selected by the radio button. Therefore, since only appropriate items are displayed as options of the analysis condition of the usage status of the selected item, it is possible to prevent the user from selecting an inappropriate condition by mistake, making it possible to appropriately analyze the usage status of the selected learning function or the like. As a result, the user can easily grasp the usage status of the selected learning function. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to select appropriate analysis conditions in accordance with the learning functions provided by the server **10**.

[0089] The controller **11** waits for reception of operation information from the first terminal **20A** via the communicator **13** indicating that an operation of selecting the analysis condition has been performed (step **S4**). When the first terminal **20A** receives the screen information of one of the screen **342** to the screen **345** transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen corresponding to the received screen information in cooperation with the browser. When an operation of selecting the analysis condition is performed by an operation via the operation unit **23A** from the displayed screen, the controller **21A** transmits operation information to the server **10** via the communication unit **25A** indicating that the operation of selecting the analysis condition has been performed. The operation information includes information indicating the selected analysis condition including the analysis unit.

[0090] When receiving the operation information from the first terminal **20A** indicating that the operation of selecting the analysis condition has been performed (step **S4**; YES), the controller **11** analyzes the usage status of the item to be analyzed based on the selected analysis condition (step **S5**). Then, the controller **11** causes the display **24A** of the first terminal **20A** to display a screen of the analysis result (step **S6**). That is, the controller **11** generates screen information for displaying the analysis result by the browser and transmits the generated screen information to the first terminal **20A** via the communicator **13**.

[0091] For example, when “Sticky Note Function” is selected as the item to be analyzed,

“Homeroom” as the analysis unit, “Homeroom 3, Grade 7” as the homeroom name to be analyzed, and “Number of Sticky Notes Created” as the display content, the controller **11** derives the number of electronic sticky notes created by each student belonging to the homeroom selected as the analysis target for each type of sticky note. For example, first, the controller **11** extracts students of the selected homeroom from the user database **124**. Next, the controller **11** derives the number of electronic sticky notes created by each extracted student for each type of sticky note based on the data stored in the operation history database **128**. Then, the controller **11** generates screen information for displaying a graph on the screen **342** by graphing the number of electronic sticky notes created by each extracted student in a different manner for each type of sticky note. When the first terminal **20A** receives the screen information of the screen **342** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **342** corresponding to the received screen information in cooperation with the browser. As shown in FIG. **12**, on the screen **342**, the graph is displayed in which the horizontal axis represents each student belonging to Homeroom 3, Grade 7, and the vertical axis represents the number of electronic sticky notes created. The graph is displayed in a different manner for each type of sticky note. For example, the graph is displayed in a color-coded manner for each type of sticky note. The teacher or the administrator can grasp what type of electronic sticky notes each student in the selected homeroom creates and uses for learning. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the usage status of the sticky note function provided by the server **10** for each homeroom.

[0092] For example, when “Sticky Note Function” is selected as the selected item, “Course” as the analysis unit, “Teacher A” as the teacher to be analyzed, and “Number of Sticky Notes Created” as the display content, the controller **11** derives the number of electronic sticky notes created by each student belonging to the course taught by Teacher A selected as the analysis target for each type of sticky note. For example, first, the controller **11** extracts students belonging to each course taught by the selected teacher from the user database **124**. Next, for each course, the controller **11** derives the number of electronic sticky notes created by each extracted student for each type of sticky note based on data stored in the operation history database **128**. Then, for each course, the controller **11** generates screen information for displaying a graph on the screen **342** by graphing the number of electronic sticky notes created by each student in a different manner for each type of sticky note. When the first terminal **20A** receives the screen information of the screen **342** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **342** corresponding to the received screen information in cooperation with the browser. On the screen **342**, the graph is displayed in which the horizontal axis represents each student belonging to the course, and the vertical axis represents the number of electronic sticky notes created. The graph is displayed in a different manner for each type of sticky note. For example, the graph is displayed in a color-coded manner for each type of sticky note. The teacher or the administrator can grasp what type of electronic sticky notes each student in the course taught by the selected teacher creates and uses for learning. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the usage status of the sticky note function provided by the server **10** for each course.

[0093] For example, when “Sticky Note Function” is selected as the selected item, “School” as the analysis unit, and “Number of Sticky Notes Created” as the display content, the controller **11** derives the transition of the total number of electronic sticky notes created during a designated period in the school to which the logged-in user belongs. The school to which the logged-in user belongs is the school associated with the user in the user database **124**. For example, first, the controller **11** extracts students belonging to the school from the user database **124**. Next, based on data stored in the operation history database **128**, the controller **11** calculates the total (cumulative)

number of electronic sticky notes created by the extracted students from the start of the designated period for each predetermined period in the designated period. Then, the controller **11** generates screen information for displaying a graph on the screen **342** by arranging the calculated total numbers in chronological order. When the first terminal **20A** receives the screen information of the screen **342** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **342** corresponding to the received screen information in cooperation with the browser. As shown in FIG. **13**, on the screen **342**, the graph is displayed in which the horizontal axis represents the period, and the vertical axis represents the (total) number of electronic sticky notes created in the entire school. The administrator can grasp the transition of the creation status of the electronic sticky notes in the entire school during the designated period. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the usage status of the sticky note function provided by the server **10** in the entire school.

[0094] For example, when “Electronic Dictionary Function” is selected as the selected item, “Homeroom” as the analysis unit, “Homeroom 1, Grade 7” as the homeroom name to be analyzed, and “Number of Dictionary Contents Searches” as the display content, the controller **11** derives the number of electronic dictionary searches for each electronic dictionary by each student belonging to the homeroom selected as the analysis target. For example, first, the controller **11** extracts students belonging to the selected homeroom from the user database **124**. Next, the controller **11** derives the number of searches for each electronic dictionary by each extracted student based on data stored in the operation history database **128**. Then, the controller **11** generates screen information for displaying a graph on the screen **343** by graphing the number of electronic dictionary searches by each student in a different manner for each electronic dictionary. When the first terminal **20A** receives the screen information of the screen **343** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **343** corresponding to the received screen information in cooperation with the browser. As shown in FIG. **14**, on the screen **343**, the graph is displayed in which the horizontal axis represents each student belonging to Homeroom 1, Grade 7, and the vertical axis represents the number of electronic dictionary searches. The graph is displayed in a different manner for each electronic dictionary. For example, the graph is displayed in a color-coded manner for each electronic dictionary. The teacher or the administrator can grasp which dictionary each student in the selected homeroom is interested in and uses for homework and/or self-study. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the usage status of the electronic dictionary function provided by the server **10** for each homeroom.

[0095] For example, when “Electronic Dictionary Function” is selected as the selected item, “Course” as the analysis unit, “Teacher A” as the teacher to be analyzed, and “Number of Dictionary Contents Searches” as the display content, the controller **11** derives the number of electronic dictionary searches for each electronic dictionary by each student belonging to the course taught by the teacher selected as the analysis target. For example, first, the controller **11** extracts students belonging to each course taught by the selected Teacher A from the user database **124**. Next, for each course, the controller **11** derives the number of searches for each electronic dictionary by each extracted student based on data stored in the operation history database **128**. Then, for each course, the controller **11** generates screen information for displaying a graph on the screen **343** by graphing the number of electronic dictionary searches by each student in a different manner for each electronic dictionary. When the first terminal **20A** receives the screen information of the screen **343** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **343** corresponding to the received screen information in cooperation with the browser. On the screen

343, the graph is displayed in which the horizontal axis represents each student belonging to the course, and the vertical axis represents the number of electronic dictionary searches. The graph is displayed in a different manner for each electronic dictionary. For example, the graph is displayed in a color-coded manner for each electronic dictionary. By checking the graph, the teacher can check whether each student uses the dictionary as expected in light of the content of the course. The teacher can also grasp which dictionary each student taking the course is interested in and uses for homework and/or self-study. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the usage status of the electronic dictionary function provided by the server **10** for each course.

[0096] For example, when “Notebook Function” is selected as the selected item, “Homeroom” as the analysis unit, “Homeroom 1, Grade 7” as the homeroom name to be analyzed, “Operations on Notebooks” as the display content, and “By Subject” as the option, the controller **11** derives the number of operations on notebooks by each student belonging to the homeroom selected as the analysis target for each subject and for each type of operation. For example, first, the controller **11** extracts students of the selected homeroom from the user database **124**. Next, for each subject, the controller **11** derives the number of operations on notebooks by each extracted student for each type of operation based on data stored in the operation history database **128**. Then, for each student, the controller **11** generates screen information for displaying a graph on the screen **344** by graphing the number of operations on notebooks for each subject for each type of operation. When the first terminal **20A** receives the screen information of the screen **344** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **344** corresponding to the received screen information in cooperation with the browser. As shown in FIG. **15**, on the screen **344**, the graph is displayed for each student in which the vertical axis represents the subject, and the horizontal axis represents the number of operations on notebooks for each type of operation. The graph is displayed in a different manner for each type of operation on a notebook. For example, the graph is displayed in a color-coded manner for each type of operation. By checking the graph shown in FIG. **15**, the teacher or the administrator can grasp what type of operation each student actively performs on notebooks for each subject. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the usage status of the notebook function provided by the server **10** for each student and for each subject.

[0097] For example, when “Notebook Function” is selected as the selected item, “Course” as the analysis unit, “Taro Kashio” as the teacher to be analyzed, “All Courses Taught” as the course name, and “Number of Notebooks Created” as the display content, the controller **11** derives the number of notebooks created by each student belonging to each course of Mr. Taro Kashio selected as the analysis target. For example, first, the controller **11** extracts students belonging to each course of Mr. Taro Kashio from the user database **124**. Next, the controller **11** derives the number of notebooks created by each extracted student for each course of Mr. Taro Kashio based on data stored in the operation history database **128**. Then, the controller **11** generates screen information for displaying a graph on the screen **344** by graphing the number of notebooks created by each student belonging to each course of Mr. Taro Kashio. When the first terminal **20A** receives the screen information of the screen **344** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **344** corresponding to the received screen information in cooperation with the browser. As shown in FIG. **16**, on the screen **344**, a graph is displayed for each course of the selected teacher in which the horizontal axis represents each student belonging to the course and the vertical axis represents the number of notebooks created. By checking the graph shown in FIG. **16**, the administrator can confirm a teacher who is able to utilize the learning support service in a course and a teacher who is

not. In addition, the teacher can compare the number of notebooks created among his or her courses to help improve the courses. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze a user who is able to utilize the learning functions provided by the server **10** and a user who is not.

[0098] The screen **344** that is displayed when “Notebook Function” is selected as the selected item and “School” as the analysis unit is an analysis screen obtained by replacing the “Number of Electronic Sticky Notes Created” in the screen **342** with “Number of Notebooks Created”, and therefore, the description is incorporated herein. The administrator can grasp the creation status of notebooks in the entire school during a designated period.

[0099] For example, when “Number of Accesses” is selected as the selected item and “Apr. 1, 2023 to Oct. 31, 2023” as the period, the controller **11** derives the number of page views (PVs) as the number of accesses during the period for each predetermined period (for example, every half month). For example, the controller **11** aggregates the number of PVs for each predetermined period during the selected period from the access history database **129**. Then, the controller **11** generates screen information for displaying a graph on the screen **345** by graphing the number of PVs for each predetermined period during the selected period. When the first terminal **20A** receives the screen information of the screen **345** on which the graph is displayed transmitted from the server **10** via the communicator **25A**, the controller **21A** causes the display **24A** to display the screen **345** corresponding to the received screen information in cooperation with the browser. As shown in FIG. **17**, on the screen **345**, the graph is displayed in which the horizontal axis represents the period, and the vertical axis represents the number of PVs. The administrator can grasp the transition of the access status to the learning support service during the designated period. That is, according to the server **10**, the interaction process between the user and the server **10** can reliably assist the user in performing a technical task. Here, the technical task performed by the user is to analyze the access status of the learning functions provided by the server **10**. The number of accesses is not limited to the number of PVs as long as the number of accesses indicates the number of accesses to a plurality of learning functions in the school to which the user belongs. For example, the number of sessions may be used instead of the number of PVs. The graphs displayed on the screens **342** to **345** show the derived results (analysis results) of the usage status of the selected items to be analyzed.

[0100] For example, the server **10** may store score information of each of the students using the learning support service in the storage **12** or the like in association with the user ID of the student. Then, in a case where the homeroom or course is selected as the analysis unit of the usage status of the learning function, the controller **11** may cause the display **24A** of the first terminal **20A** to display the usage status of the electronic sticky note function, the notebook function, or the electronic dictionary function by each of the students belonging to the selected homeroom or course, and the score information of the student, side by side. For example, when a graph of each of the students is displayed on the screens **342** to **344**, the graph and the score information of the student may be displayed side by side. This allows the teacher or the administrator to compare scores of students who frequently use the electronic sticky note function or notebook function with scores of students who do not, thereby enabling the teacher or the administrator to grasp the correlation between the use of the electronic sticky note function or notebook function and the scores. Similarly, this allows the teacher or the administrator to compare scores of students who have searched an electronic dictionary many times with scores of students who have not, thereby enabling the teacher or the administrator to easily grasp the correlation between the number of searches of the electronic dictionary and the scores. In addition, for example, the controller **11** may notify a student to use the electronic sticky note function if the number of electronic sticky notes created or number of notebooks created (or number of operations) by the student is equal to or less than a predetermined threshold and the score of the student is also equal to or less than a

predetermined threshold. For example, the controller **11** may cause the display **24B** to display a message notifying the student to use the electronic sticky note function or the notebook function when the student logs in. This makes it possible to automatically encourage a student who rarely uses the learning functions to use the learning functions.

[0101] As described above, the controller **11** of the server **10** acquires the creation history information of electronic sticky notes created by the learner via the second terminal **20B**. When the school is designated as the analysis unit of the usage status of the electronic sticky note function by the learning instructor or the administrator via the first terminal **20A**, the controller **11** derives the total number of electronic sticky notes created by learners of the same school as the school associated with the learning instructor or the administrator based on the creation history information of the electronic sticky notes, and causes the first terminal **20A** to display the derived result. This allows the learning instructor and the administrator to easily grasp the usage status of the electronic sticky note function in the entire school.

[0102] For example, as a result of deriving the total number of electronic sticky notes created in the entire school, the controller **11** causes the first terminal **20A** to display a graph in which the horizontal axis represents the period, and the vertical axis represents the total number of electronic sticky notes created. This allows the learning instructor or the administrator to easily grasp the transition of the total number of electronic sticky notes created in the entire school in a predetermined period.

[0103] In addition, for example, when the homeroom or the course is designated as the analysis unit by the learning instructor or the administrator via the first terminal **20A**, the controller **11** derives the number of electronic sticky notes created by each learner belonging to the designated homeroom or course based on the creation history information and causes the first terminal **20A** to display the derived result. This allows the learning instructor or the administrator to easily grasp the number of sticky notes created by each learner belonging to the designated homeroom or course.

[0104] Further, for example, the controller **11** causes the first terminal **20A** to display a graph in which the horizontal axis represents a learner belonging to the designated homeroom or a learner belonging to the designated course, and the vertical axis represents the number of electronic sticky notes created, as the derived result. This allows the learning instructor or the administrator to more easily grasp the number of sticky notes created by each learner belonging to the designated homeroom or course.

[0105] There is a plurality of types of electronic sticky notes, and the controller **11** derives the number of electronic sticky notes created by each learner for each type of electronic sticky note. This allows the learning instructor or the administrator to easily grasp the number of electronic sticky notes created by each learner belonging to the designated homeroom or course for each type of electronic sticky note.

[0106] The controller **11** causes the first terminal to display a graph in which the horizontal axis represents a learner belonging to the designated homeroom or a learner belonging to the designated course, and the vertical axis represents the number of electronic sticky notes created, showing the number of electronic sticky notes created in a different manner for each type of electronic sticky note. This allows the learning instructor or the administrator to more easily grasp the number of electronic sticky notes created by each learner belonging to the designated homeroom or course for each type of electronic sticky note.

[0107] In a case where a plurality of electronic sticky notes is created using a plurality of different electronic dictionaries in a single course, the controller **11** acquires the creation history information of the electronic sticky notes created in each of the plurality of electronic dictionaries and adds up the number of electronic sticky notes created in each of the plurality of electronic dictionaries to derive the total number. Therefore, when a plurality of electronic sticky notes is created using a plurality of electronic dictionaries in a single course, the number of electronic sticky notes created in each of the plurality of electronic dictionaries can be counted towards the total number of

electronic sticky notes created.

[0108] The above embodiment describes a suitable example of the information processing apparatus, the information processing system, and the recording medium according to the present disclosure. This is not intended to limit the present invention.

[0109] In addition, for example, in the above description, as a computer readable medium storing the programs to perform the processes of the present invention, an HDD, a nonvolatile semiconductor memory or the like is used. The computer readable medium, however, is not limited to this example. The computer readable medium may be a portable recording medium such as a CD-ROM. Further, as a medium to provide data of the programs via a communication line, a carrier wave may be used.

Claims

1. An information processing apparatus that is configured to be communicably connected to a first terminal configured to be used by a learning instructor or an administrator and a plurality of second terminals configured to be used by a plurality of learners, the plurality of second terminals each including an electronic sticky note function for creating an electronic sticky note related to learning, comprising a processor that: acquires creation history information of the electronic sticky note created by the plurality of learners via the plurality of second terminals; and in response to a school being designated as an analysis unit of usage status of the electronic sticky note function by the learning instructor or the administrator via the first terminal, derives a total number of the electronic sticky note created by the plurality of learners of a same school as a school associated with the learning instructor or the administrator based on the creation history information, and causes the first terminal to display a derived result.
2. The information processing apparatus according to claim 1, wherein, as the derived result, the processor causes the first terminal to display a graph in which a horizontal axis represents a period and a vertical axis represents the total number of the electronic sticky note created.
3. The information processing apparatus according to claim 1, wherein in response to a homeroom or course being designated as the analysis unit by the learning instructor or the administrator via the first terminal, the processor derives numbers of the electronic sticky note created by the respective plurality of learners belonging to the designated homeroom or course based on the creation history information, and causes the first terminal to display a derived result.
4. The information processing apparatus according to claim 3, wherein the electronic sticky note includes a plurality of types of electronic sticky notes, and the processor derives the numbers of the electronic sticky note created by the respective plurality of learners with respect to each type of the plurality of types of electronic sticky notes.
5. The information processing apparatus according to claim 3, wherein, as the derived result, the processor causes the first terminal to display a graph in which a horizontal axis represents a learner belonging to the designated homeroom or course and a vertical axis represents a number of the electronic sticky note created by the learner.
6. The information processing apparatus according to claim 4, wherein, as the derived result, the processor causes the first terminal to display a graph in which a horizontal axis represents a learner belonging to the designated homeroom or course and a vertical axis represents a number of the electronic sticky note created by the learner, showing the numbers of the electronic sticky note created in a different manner with respect to each type of the plurality of types of electronic sticky notes.
7. The information processing apparatus according to claim 1, wherein the electronic sticky note includes an electronic sticky note created using an electronic dictionary, and, when the electronic sticky note includes a plurality of electronic sticky notes created using a plurality of electronic dictionaries in a single course, the processor: acquires creation history information of the electronic

sticky note created using each of the plurality of electronic dictionaries; and adds up numbers of the electronic sticky note created using the respective plurality of electronic dictionaries to derive the total number of the electronic sticky note.

8. An information processing system comprising: the information processing apparatus according to claim 1; and the first terminal and the plurality of second terminals communicably connected to the information processing apparatus.

9. A non-transitory computer-readable recording medium storing a program executed by a computer of an information processing apparatus, wherein: the information processing apparatus is configured to be communicably connected to a first terminal configured to be used by a learning instructor or an administrator and a plurality of second terminals configured to be used by a plurality of learners, the plurality of second terminals each including an electronic sticky note function for creating an electronic sticky note related to learning; and the program causes the computer to: acquire creation history information of the electronic sticky note created by the plurality of learners via the plurality of second terminals; in response to a school being designated as an analysis unit of usage status of the electronic sticky note function by the learning instructor or the administrator via the first terminal, derives a total number of the electronic sticky note created by the plurality of learners of a same school as a school associated with the learning instructor or the administrator based on the creation history information, and causes the first terminal to display a derived result.
